Checklist Starting System and example calculation

TDA/04/19

MTU

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Note:

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Documents released: **Checklist Starting System and example calculation**

Version Number: V1.6

Location and Date of Issue: Friedrichshafen, 2021-03-25

Author:

Name	Department	Role	Date	Signature
Angelo Germann Stephan Schneider Matthias Dorner Eduard Erbert	TDAP EECE TSFL TDAP	Application Engineers		

Verified:

Name	Department	Role	Date	Signature
Muecke Marcus	TDAP	Senior Manager		

Approved:

Name	Department	Role	Date	Signature
Armin Fuerderer	EEC	Director		
Michael Koliwer	TSF	Director		

Revision Sheet

Rev.	Date	Chapter/Page	Scope of the Revision	Author
1.1	2017-11-30		Various amendments	Smithback, Demharter
1.2	2018-12-11		First issue	Schneider
1.3	2019-04-10		Second issue	Germann
1.4	2019-04-26		Third issue	Germann
1.5	2020-03-24		Revision of chapters and content	Erbert, Dorner
1.6	2021-03-25		Corrections	Erbert, Dorner

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List of Abbreviations

Abbreviation	Definition
ECU	Engine Control Unit
h	hour (time unit)
Hz	Hertz (frequency unit)
I	Current [A]
IEC	International Electrotechnical Commission
K	Kelvin (temperature unit)
M	Starter motor
R_G	Resistance of the battery and the cabling (given on the Starter datasheet) → ($R_G = R_B + R_L$)
R_B	Internal resistance of the battery (system) → ($R_B = R_{Br} + R_{Bat}$)
R_{Bridge}	Resistance of the battery bridge cabling
R_{B_i}	Internal resistance of the 12V batterie(s)
$R_{C,plus}$	Resistance of the plus cabling
$R_{C,minus}$	Resistance of the minus cabling
R_L	Resistance of the cabling → ($R_L = Ri- + Ri+$)
R_M	Internal resistance of the starter motor
T	Temperature [°C]
T_{EMin}	Minimum environmental temperature
T_{EMax}	Maximum environmental temperature
U	Voltage [V]
U_0	Open circuit voltage
Ω	Ohm

Table 1: List of Abbreviations

1 Scope

This document explains the steps enabling to choose the correct battery and cables according to the starter used. In order to confirm the size of battery cables, it is necessary to compare the total resistance of the battery and cables to the value R_G which can be found in the respective equipment manufacturer data.

To define the minimum battery capacity it is necessary to consider the starting current for each starter (equipment manufacturer data) as well as the starting time and number of start attempts (defined by application norms). If an uninterruptible power supply for other components has also to be provided by the batteries, their power consumption and the time to be bridged must be taken into account.

This information sheet has been realized by considering an operating temperature as equal to **20** °C for the calculation.

2 References

No.	Document Number	Issue	Title
[1]	EN 61660-2	1997	Short-circuit currents in d.c. auxiliary installations in power plants and substations – Part 2: Calculation of effects (IEC 61660-2:1997)
[2]	ISO 8528-12 (Part 8)	1997(E)	Reciprocating internal combustion engine driven alternating current generating sets -- Part 12: Emergency power supply to safety services
[3]	NFPA® 20	2019 Reference: 10.10.3.5 TIA 19-1	Standard for the Installation of Stationary Pumps for Fire Protection
[4]	DIN 6280-13	1994-12	Generating sets - Reciprocating internal combustion engines driven generating sets - Part 13: For emergency power supply in hospitals and public buildings
[5]	DIN EN 60865-1	2012-09	Short-circuit currents - Calculation of effects - Part 1: Definitions and calculation methods

Table 2: List of References

3 List of Appendices

No.	Issue	Title	Pages
[1]	2005-06	Datasheet OPTIMA® YellowTop S 5,5	2
[2]	2017-02	Datasheet H01N2-D 1 x 3/0 AWG	1
[3]	2016-02	Datasheet SAFT SPH 170	2
[4]	2015-04	Datasheet HELUKABEL MULTISPEED 600-PUR-J	1
[5]	2019-05	Datasheet LAPP H07RN-F, enhanced version	1
[6]	2017-09	Datasheet RADOX 4GKW-AX 1800V M	4

Table 3: List of Appendices

4 Procedure Checklist

For correct starting system design is it advisable to work through the following points.

a. Definition of the starting system
b. Battery Selection
   - Information from the equipment manufacturer datasheet
   - Applicable norm and standard
   - Calculation of the required battery capacity
   - Checking of the short circuit current
   - Selection of the battery configuration
c. Designing of the cables
   - Calculation of the equivalent cabling resistance
   - Selection of the standard cross section according to the expected current
   - Check of the temperature rise of cables after the starting cycle
d. Assessment of the selected components
e. Review of the selected components by measurement

5 Starting System Definitions

MTU offers a variety of starter packages to meet the needs of customer applications. In addition to standard and oversized starter options, MTU offers a redundant starter system that provides seamless operation in case of a starter failure. This optional starting system is both factory-configurable and field-installable.

5.1 Correct use of the Starters

For the correct use of the starters it is important and necessary to comply with the limit values set by the manu-facturer. These values are operating voltage, rated short circuit current, power consumption (at an engine speed of 100 rpm) and the internal resistance of power supply. All this information are available on the Business Portal or TEN-database.

Example for a ***Prestolite M105R*** starter:

Parameters*	**Units**
Power output	15 kW
Voltage	24 V
Short circuit current	3000 A
Starting current (at 100 rpm speed of the engine)	1400 A
Resistance of battery and cabling (R_G)	4,5 mΩ

*at 20°C

6 Definition of the starting system

First, the starting system has to be defined according to the engine configuration and the use of the customer:

- The type of starter configuration (single, parallel or redundant)
- The norm and regulation the customer needs to fulfill
- Desired or needed type of batteries

Typically is one starter connected to one battery pack (see Figure 6.1). If the batteries are able to provide the short-circuit current for two starters, it is also possible to connect two starters to one batterie pack (see Figure 6.2).

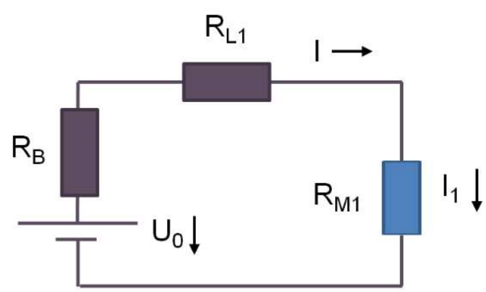

Figure 6.1: Configuration as single starter

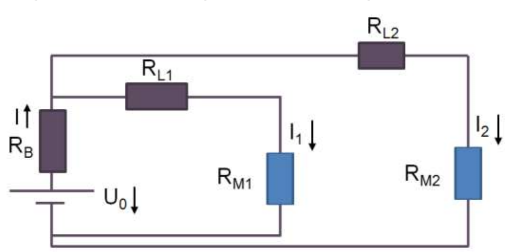

Figure 6.2: Configuration as parallel starters

In the second configuration, the length of the cabling and the current in the second branch is different than the first one. This must be taken into account in the calculation of the cross section of the cables. However, to avoid possible errors, it is recommended to use the same length of cabling.

INFORMATION

- If you use other starters please consider the data provided by the manufacturer
- It is crucial to size the battery configuration and cabling according to the guidelines given by the manufacturer of the equipment.
- **Mentioned data are field experience of MTU based on 20 years**